

Chapter 7

A Simple Risk Management Model That Works

“We find that by and large, these models [risk-management, or market-timing models] have predicted poorly both in-sample (IS) and out-of-sample (OOS) for 30 years...”

—Goyal and Welch, Comprehensive Look at the Empirical Performance of Equity Premium Prediction

Warning: The following chapter is highly “quant.” If you enjoy programming, engineering, mathematical formulas, or took your prom date to the planetarium, this chapter is for you. On the other hand, if you are a newbie to the DIY universe, never fear. We have split this chapter into two sections. The first section gives you the key takeaways. Newbies and quants alike will enjoy this section. The second section drills down into the academic research behind the key takeaways. Good reading, but certainly not required to get the basics. Let's press onward!

Asset allocation is all well and good, but many investors, ourselves included, strive to manage risk to the utmost. If you have a great asset allocation model that drops by 40 percent when the market drops by 60 percent, are you smiling saying: “Isn't that great? I am beating the market by 20 percent!” Of course not. The art of preventing significant drawdowns, better known as risk management, is a well-studied field in high finance. Like asset allocation, there are numerous hypotheses and competing arguments as to what approach works.

We have examined hundreds of risk-management, or *market-timing*, platforms over the years. You name it, we've probably seen it.

There are five broad “buckets” of risk management techniques:

1. Fundamental indicators—assess macroeconomic factors
2. Technical indicators—assess past prices, stock metrics